

FYKE SIMON V. TONEL

Metro Manila, Philippines | 0961-575-4156 | fyketonel22@protonmail.com | fykelabs.vercel.app

Education

STI College Caloocan

2022 – Present

Bachelor of Science in Computer Science

Caloocan City, Metro Manila

- Current Cumulative GWA: 1.27
- Best Thesis, Bachelor of Science in Computer Science (Campus-wide – ICTE Symposium), 2024-25
- Best Methods of Research Project, Bachelor of Science in Computer Science (Campus-wide – ICTE Symposium), 2024–25
- Entrant, Codefest (Local – Tagisan ng Talino), 2024–25
- President’s Honor (STI Academic Center – Academic Honors), 2022 – Present
- Alumni Scholar (Awarded for High Honors in Senior High; 50% scholarship in college)

STI College Caloocan

2020 – 2022

Senior High School (Information & Communications Technology)

Caloocan City, Metro Manila

- Graduated with High Honors
- Completion of Computer Programming – Java and .NET Technology (IT in Mobile App & Web Development)

Lakan Dula High School

2016 – 2020

Junior High School

Manila City, Metro Manila

- Graduated with Honors

Projects

Flick Player: Music Player with Custom UAC 2.0

January 2026

Flutter, Dart, Rust

Lead Developer

- Engineered a high-fidelity audio engine using **Rust**, implementing a custom **USB Audio Class (UAC) 2.0 driver** to bypass the Android OS mixer (AudioFlinger) for bit-perfect, lossless playback.
- Architected a robust **Flutter-to-Rust bridge** via **Dart FFI (Foreign Function Interface)**, ensuring thread-safe communication between the reactive UI and the low-latency audio processing background service.
- Optimized the decoding pipeline to support high-resolution formats (FLAC, DSD, WAV) with efficient memory allocation, ensuring stable asynchronous USB transfer rates and minimal CPU overhead.
- Designed an immersive, audiophile-centric interface in **Flutter**, featuring real-time audio visualization and metadata management, while maintaining 60 FPS performance during heavy signal processing.

Revo: Productivity & Study Companion Application

December 2025

React, Next.js, TypeScript, Tailwind CSS, Supabase, PostgreSQL, Spotify Web API

Lead Developer

- Developed a productivity application featuring an integrated calendar, note-taking system, and study-focused timer.
- Implemented multiple learning techniques including Pomodoro, 52/17, Deep Work, Short Burs, and custom study cycle configuration.
- Integrated Spotify Web API to allow users to sync and stream their personal music libraries within the app.

Locker: Advanced Media Hiding & Security Application

November 2025

Flutter, Dart

Lead Developer

- Developed a secure media-hiding application allowing users to conceal pictures, videos, and documents with AES encryption.

- Implemented a decoy file system to thwart unauthorized access attempts and brute-force attacks.
- Designed an auto-kill feature to terminate the app when it moves to the background, enhancing privacy protection.

PASADA: Ride-Hailing & Fleet Management Ecosystem

December 2024 – November 2025

Flutter, TypeScript (Bun/Deno), React, Supabase, Google Cloud Platform

Lead Full-Stack Developer

- Architected a modernized public transport solution interconnecting four distinct applications: Pasada Passenger, Pasada Driver, Pasada Admin, and the Pasada Official Website.
- Engineered a custom analytics pipeline for the Admin Dashboard that provides 7-day predictive forecasts for both cooperative booking frequency and the route traffic.
- Integrated the Gemini API within both Passenger and Admin applications to interpret complex, in-house predictive and real-time traffic data into easily digestible, actionable, summaries for users and fleet managers.
- Developed the Passenger and maintained the Driver mobile applications using Flutter, featuring real-time, Google Maps-sourced traffic analytics for optimized routing and planning.
- Built the official website using Next.js and Tailwind CSS, featuring a live public ride-tracker implemented with MapLibre GL JS and OpenStreetMaps.

Experience

Back-End Developer (Intern)

January 2026 – February 2026

Inspire Holdings Inc.

- Serves as the Technical Lead for *Inspire Wallet* and *Admin 2.0*, a FinTech investment platform, overseeing the full backend lifecycle and architectural decisions.
- Architected a scalable backend infrastructure using NestJS and AWS, implementing load balancing, rate limiting, and local/remote caching via Redis to ensure high availability.
- Engineered the payment gateway integration for versions 1.0 and 2.0 using the UnionBank API, ensuring secure and seamless financial transactions.
- Optimized database performance and management using Supabase and ORMs, while handling transactional emails via Resend.
- Wrote comprehensive unit tests to validate core logic and security features, and collaborated cross-functionally to ensure UI/UX quality standards were met.
- Defined and implemented the company's first proper SDLC and a collaborative QA testing protocol involving cross-team validation to reduce bugs in production.

Skills

Programming Languages: TypeScript, JavaScript, Rust (Systems), Java, Dart, SQL, HTML, CSS, Bash/PowerShell.

Backend & Cloud: Node.js, NestJS, Express.js, Bun, Deno, AWS, GCP, Vercel, Cloudflare, Fly.io, Docker.

Database & Caching: PostgreSQL, MySQL, SQLite, Supabase, Redis, Firebase, QuestDB (Time-series).

Frontend & Mobile: Flutter, React, Next.js, Remix, Three.js, Tailwind CSS, MapLibre GL JS.

AI & LLM Tools: Ollama, LM Studio, HuggingFace, Gemini API.

Testing & Tooling: Jest, Postman, pnpm/npm, Git (GitHub/GitLab), Jira, Mermaid.js.

Environment: Linux (Arch Linux), Neovim, Cursor, Zed, Google Antigravity, Android Studio, Figma, Eraser.io, Draw.io.

Certifications

Learn Dart: First Step to Flutter

July 2024

Educative.io

Java Fundamentals

June 2023

Oracle